Project: Compliance Track – Internal Policy Compliance & Audit Management System

Compliance Track is an integrated corporate compliance management system that centralizes user access control, policy creation and distribution, employee acknowledgements, audit task tracking, assessments, and real-time notifications. It ensures that the right employees receive the right policies, complete mandatory acknowledgements and quizzes, and carry out assigned audit tasks while giving managers and admins full visibility into compliance progress through analytics and reports. The system automates end-to-end compliance workflows—from policy assignment and quiz evaluation to audit checklist execution—making corporate governance more structured, transparent, and efficient across the organization.

USER MANAGEMENT MODULE

1) MODULE DESCRIPTION

The **User Management Module** is responsible for handling all user-related operations within the Compliance Track system. It enables secure creation, authentication, and role-based authorization of users such as Employees, Managers, and Administrators. This module forms the foundation of the system's access control, ensuring that only valid and active users can log in and that each user can access only those features permitted for their assigned role.

2) KEY OBJECTIVES

The primary purpose of the User Management Module is to:

- Maintain a centralized repository of all system users
- Provide secure login and authentication mechanisms
- Enforce role-based access control (RBAC) across the application
- Support user lifecycle management handled by the Admin role
- Ensure only authenticated and authorized users interact with other modules

3) IN SCOPE

3.1 User Registration

- Users can create new user accounts.
- Enters essential attributes while registering:
  - Name
  - Email (acts as username)
  - Role (Employee / Manager / Admin)
  - Department

    - Employee Type

3.2 User Authentication (Login)

- Common login endpoint for all roles.
- Validates username and password.
- Rejects login for **inactive** users.
- Supports password initialization using a limited-scope token.

3.3 Role-Based Authorization

- After login, the backend includes user role information in the token.
- The frontend redirects users to different dashboards based on the role:
    - Employee Dashboard
    - Manager Dashboard
    - Admin Dashboard
- Backend enforces access rules for all protected APIs (RBAC).

3.4 User Status Management

- An inactive user cannot log in to the system.
- Admin can deactivate or reactivate any user.
- Soft deletes ensure audit history remains intact.

3.5 Session and Security Handling

- Authentication via JWT (or equivalent secure token).
- Role included inside token claims for authorization.
- Passwords stored using secure hashing algorithms.
- All endpoints protected using role-based checks on backend.

4) DATA MODEL

A) USER

- userId
- name
- email
- role
- departmentId
- employeeType
- status
- password
- createdAt
- managerId

5. ACTORS AND USECASES

5.1. User (Employee / Manager / Admin — Shared actor for registration/login)

A system user who interacts with:

- Registration
- Login

5.2. Admin

An Admin has elevated privileges and interacts with:

- Managing user status (activate/deactivate)
- Updating user details
- Viewing all users

6. WORKFLOW

6.1 Registration Workflow

User enter details → Encrypt password→ User stored

6.2 Login Workflow

User submits login
→ Verify user status
→ Validate password
→ Generate JWT and redirect to role-based dashboard

6.3 Role-Based Navigation

Role = EMPLOYEE → Employee Dashboard
Role = MANAGER → Manager Dashboard
Role = ADMIN → Admin Dashboard

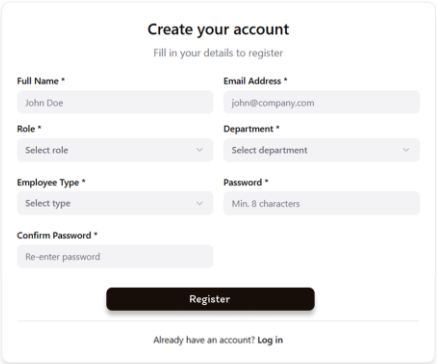
7. ENDPOINTS

Sl. No.	Methods	URLs	Description
1	POST	/auth/register	Register user
2	POST	/auth/login	User Login
3	GET	/admin/users	List all users
4	GET	/admin/users/{id}	Get user by ID

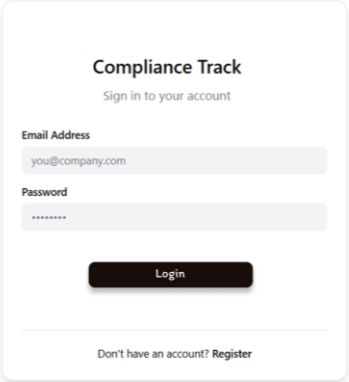
5	PUT	/admin/users/{id}	Update user details
6	PATCH	/admin/users/{id}/status	Activate/Deactivate user

8. SCREENS

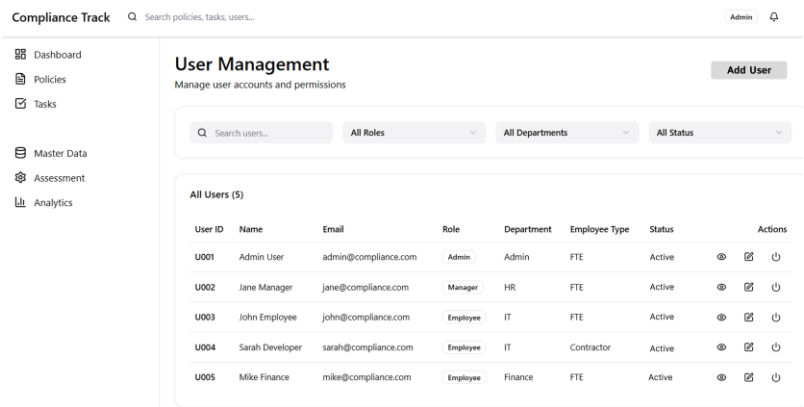
8.1 Registration Screen:



8.2 Login Screen:



8.3 User Management Screen:



POLICY MANAGEMENT MODULE

1)MODULE DESCRIPTION

The Policy Management Module provides a centralized system for creating, updating, organizing, and maintaining company policies. Administrators can define policy details such as category, status, due days, and task requirements. Policies can be mapped to specific employee types and departments to ensure correct visibility. Managers and employees can view only the policies that are relevant to their roles. The module helps maintain structured, accessible, and well-organized policy governance across the organization.

2) KEY OBJECTIVES

1. **Centralized Policy CRUD (Admin only)**
   → Create, read, update, and delete policies to maintain accurate and up-to-date policy information.
2. **Search and filter policies** based on category, status, employee types, or departments for quick retrieval.
3. **Mark audit-enabled policies:**
   → HasTask = true → Policy supports audit checklists or tasks.
   → HasTask = false → Policy is informational only.
4. **Clean categorization:**
   → Category and Department mappings ensure that employees and managers view only the policies that apply to them.

3)IN SCOPE

- **Creating, updating, and deleting policies** within the system.
- **Determining policy applicability** based on selected employee types, departments, and managerial visibility settings.
- **Allowing Administrators to view all users** who have been assigned a specific policy for oversight and management.

4) DATA MODEL

POLICY:

- PolicyID (PK)
- Title
- Description

- EffectiveDate
- Status
- HasTask (boolean)
- DueDays
- PolicyCategoryID
- EmpTypeID *(via join table)*
- DepartmentType *(via join table)*
- QuizID

JOIN TABLES

policies_employee_type

- policy_id
- emp_type_id

policies_department_type

- policy_id
- dep_id

5) ACTORS & USE CASES

5.1 Admin

- Create, edit, delete, or archive a policy.
- Assign a policy to employee types, department types, and managers in one action.
- Enable or disable the HasTask option based on whether the policy includes tasks.
- Assign a policy to specific employees when needed.

5.2 Manager & Employee

- Views policies that apply to their department or policies directly assigned to them.
- Cannot create, edit, or delete any policy.

6)WORKFLOW

- **Admin Creates Policy**
  → Enters policy details → Sets category, effective date, HasTask → Selects employee types and departments → Saves the policy.

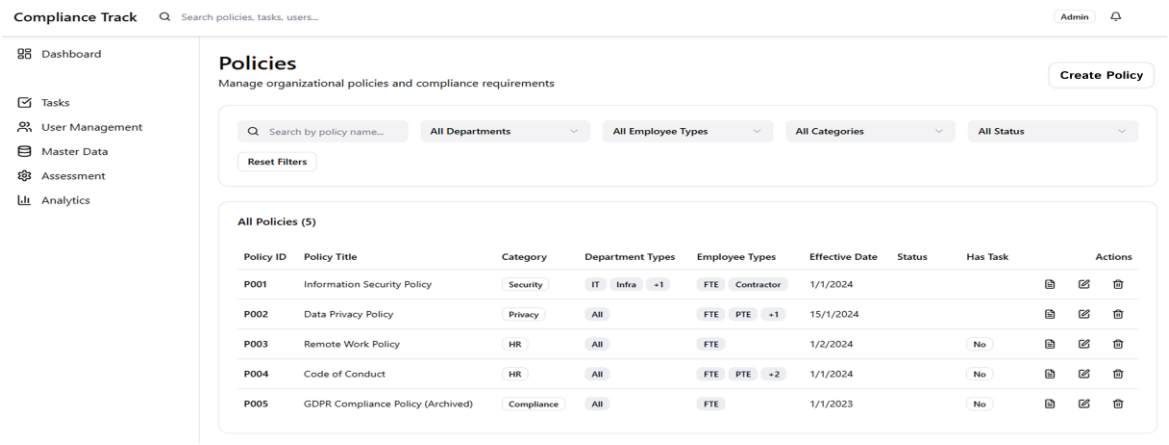
- **Admin Assigns Policy**
  → System identifies all employees who match the selected employee types and departments → Assigns the policy to those users.
- **Admin Updates Policy**
  → Edits policy details or applicability → System recalculates matching users → Adds assignments for any new users if required.
- **Admin Deletes Policy**
  → Policy is removed from the system.

7) ENDPOINTS

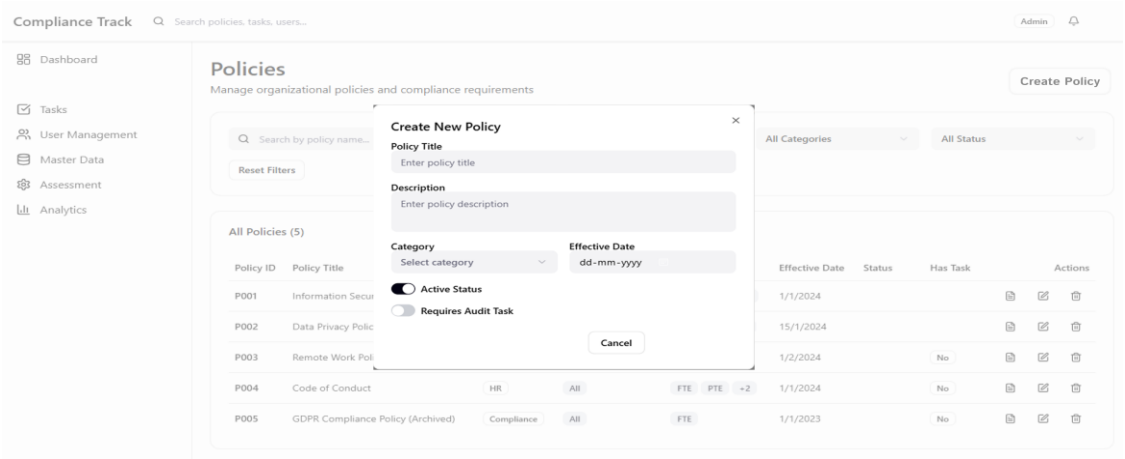
S.No	METHOD	URL	DESCRIPTION
1	POST	/admin/policy/createPolicy	Creates a new policy with details.
2	PUT	/admin/policy/{policyId}/editPolicy	Updates the policy's details.
3	DELETE	/admin/policy/{policyId}/deletePolicy	Removes a policy from the system.
4	GET	/admin/policy/viewAllPolicies	Returns the full list of policies.
5	GET	/admin/policy/searchPolicy?keyword={keyword}	Searches policies by keyword in title or description.
6	GET	/admin/policy/viewAllPolicies/filterPolicies ?policyStatus={status} &categoryIds={1,2,...} &employeeTypeIds={1,2,...} &DeptIds={1,2,...}	Policy filter/search By admin
7	GET	/admin/policy/{policyId}/viewPolicy	Policies directly assigned to the manager

8) SCREENS

8.1 Policy List:



8.2 Create Policy



POLICY ASSESSMENT MODULE

1) MODULE DESCRIPTION

Ensure that when a user is assigned a policy, a corresponding quiz is auto-assigned. The user must complete the quiz with a score ≥ **75%** to be considered **compliant** with that policy.

2)KEY OBJECTIVES

- Question bank per policy (MCQ).
- Versioned quizzes aligned to policy versions.
- Configurable pass threshold (default 75%).

3)IN SCOPE

- Auto-provision quizzes on policy assignment.
- Enforce a pass threshold (configurable, default **75%**).
- Track scores.
- Compliance for policy is enabled once the assessment is completed with required score.

4)DATA MODEL

A) Quiz_Assignment

- quizAssignmnetId
- userId
- quizId FK
- quizScore

B) Quiz

- quizId
- quizTitle
- policyId

C) Question

- questionId
- questionText
- options
- correctOption
- points
- quizId

5)ACTORS AND USE CASES

5.1 Employee and Manager:

- Views assigned quizzes for the assigned policies, attempts quiz and view the score.

5.2 Admin:

- Creates assessment, updates and deletes the assessments based on the policies

6) WORKFLOW

A) Policy Assignment → Auto-Quiz

- Admin assigns Policy (or auto via rule/group).
- **Assignment Service** creates Assignment for each user targeting **current PolicyVersion**.
- Each policy consists of set of 5 Multiple choice questions.

B) Quiz Attempt & Scoring

1. User opens quiz.
2. User submits answers → **Assessment Engine**:
   a. Compares selections with correct options.
   b. Computes total points → score= (earned/totalPossible) * 100.
   c. passed = score_pct ≥ pass_threshold_pct.
3. If **passed**:
   a. **Compliance Service** marks ComplianceStatus.status = compliant.
   b. Assessment status → completed.
   c. Ready for the policy compliance
4. If **failed**:
   a. Record failed attempt keep retake the assessment till test is passed.

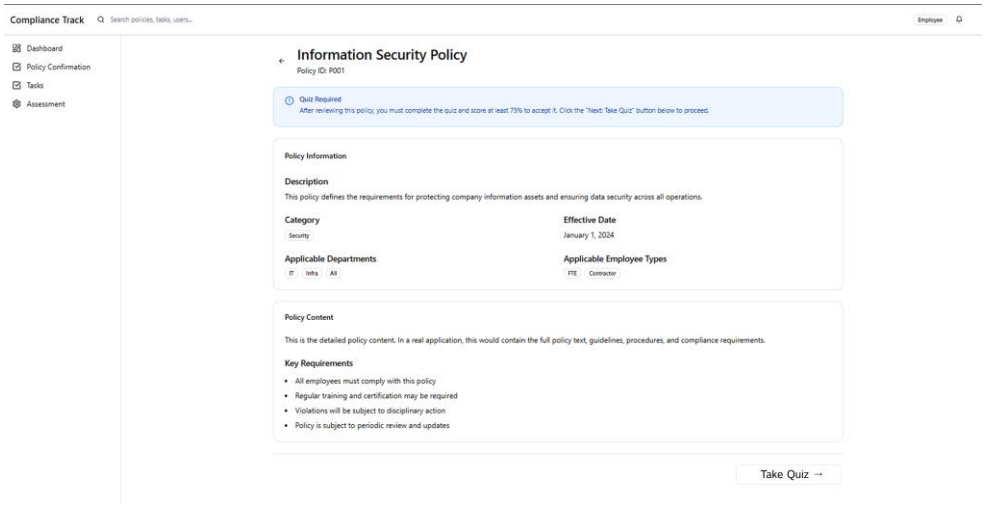
7) ENDPOINTS

Sr. No	REST Method	URL	Description

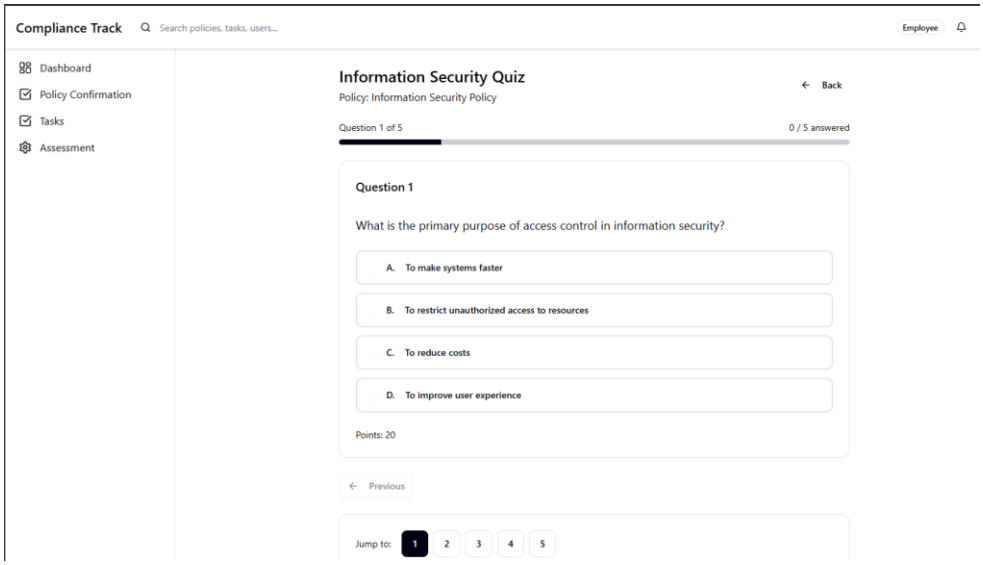
1	POST	/api/quizzes	Create a quiz for a policy (inserts into **Quiz** table)
2	GET	/api/quizzes/{quizId}	Get quiz details by quizId (from **Quiz** table)
3	GET	/api/policies/{policyId}/quiz	Get quiz linked to a policy (uses **Quiz.policyId**)
4	POST	/api/questions	Add a question to a quiz (inserts into **Question** table)
5	GET	/api/quizzes/{quizId}/questions	Fetch all questions for a quiz (from **Question** table)
6	POST	/api/quiz-assignments	Assign quiz to user when policy is assigned (inserts into **QuizAssignment**)
7	GET	/api/users/{userId}/quiz-assignments	Get all assigned quizzes for a user (from **QuizAssignment**)
8	GET	/api/quiz-assignments/{quizAssignmentId}	Get assignment details (quizId, score, status)
9	POST	/api/quiz-assignments/{quizAssignmentId}/start	Mark quiz as started (update **QuizAssignment**)
10	POST	/api/quiz-assignments/{quizAssignmentId}/submit	Submit quiz answers & calculate score (update **quizScore**)
11	GET	/api/users/{userId}/policies/{policyId}/compliance	Check compliance status (quizScore ≥ 75)
12	GET	/api/reports/quiz-compliance	Get compliance report across users/policies

8) SCREENS

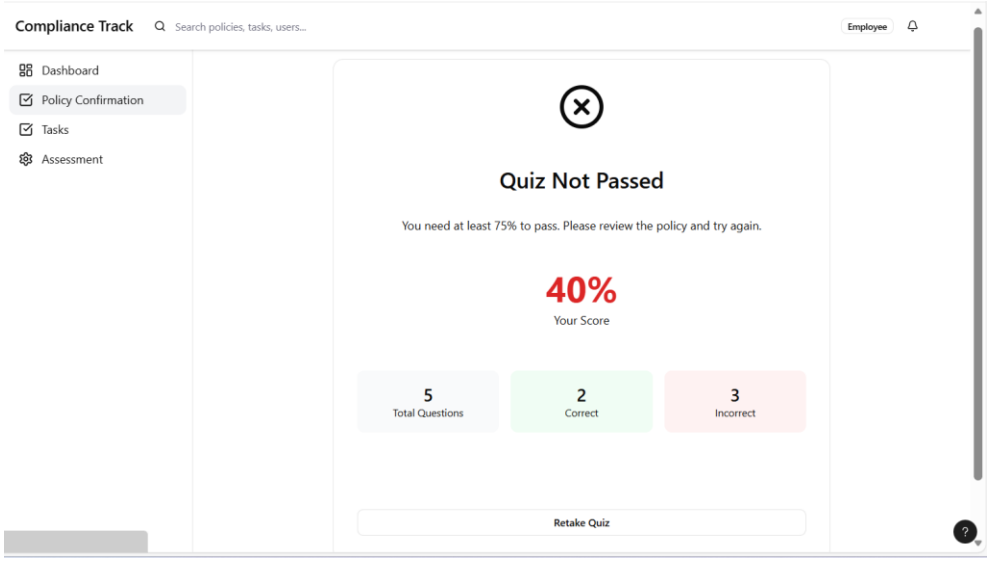
8.1 Entry point for quiz



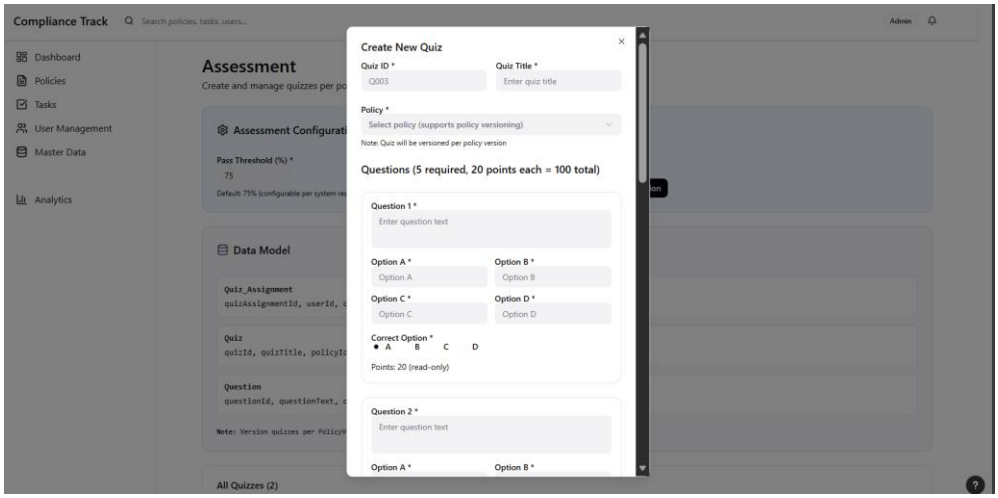
8.2 Employee and Manager Quiz page

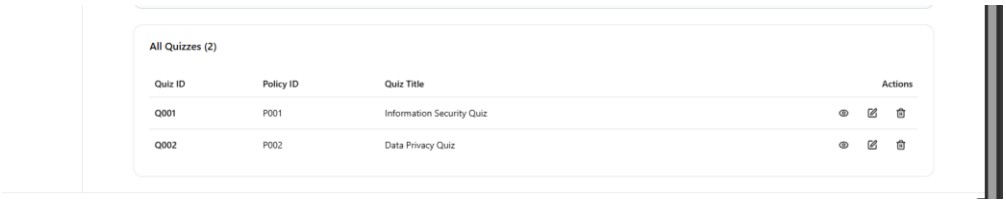


8.3 Quiz result page



8.4 Quiz Crud Screens





POLICY COMPLIANCE MODULE

1) MODULE DESCRIPTION

The **Policy Compliance Module** allows administrators, managers, and employees to track and manage policy compliance across the organization. Administrators can view all user compliance records, apply filters, and search for specific users to monitor overall policy adherence. Managers can view policies assigned to them, accept policies, and also access the compliance records of employees under their supervision. Employees and managers can review their assigned policies along with details such as category, due date, and status, and acknowledge them when required. The module provides robust filtering and viewing capabilities to support efficient compliance monitoring and timely policy completion.

2) KEY OBJECTIVES

- To display all policies assigned to employees and managers in a clear and organized format.
- To allow employees and managers to acknowledge assigned policies through an acceptance action.
- To enable efficient filtering of compliance records for quick identification and review.
- To accurately update and store policy acknowledgement status within the system's database.

3) IN SCOPE

- Displaying all policies assigned to employees and managers.
- Providing filtering options to narrow down compliance records based on defined criteria.
- Enabling employees and managers to acknowledge and accept assigned policies.
- Persisting and updating policy acknowledgement status in the system.
- Allowing administrators to view all users' compliance records.
- Allowing administrators to filter compliance records and search for specific users.

4)DATA MODEL

COMPLIANCE_RECORD

- **record_id**
- **policy_id**
- **user_id**
- **assigned_date**
- **confirmation_date**

- **due_date**
- **status**
- **task_id**

5) ACTORS AND USE CASES

a)Admin

- View all users' compliance records
- Filter compliance records by category
- Filter compliance records by due date
- Filter compliance records by status
- Search users by name
- Assign policies to eligible users
- Delete an assigned policy record

b)Manager

- View all policies assigned to them
- Filter policies by category
- Filter policies by due date
- Filter policies by status
- Accept (acknowledge) a policy
- View compliance records of employees under their supervision

c)Employee

- **View all assigned policies**
- **Filter policies by category**
- **Filter policies by due date**
- **Filter policies by status**
- **Accept (acknowledge) a policy**

6)WORKFLOW

1. Admin (Compliance Management)

a. Assigns newly created policies to all eligible users.
b. Views all users' compliance records in the dashboard.
c. Applies filters (category, due date, status) and searches users by name.
d. Manages compliance records (e.g., delete an assigned policy record).

2. Manager (Team & Self Compliance)

a. Views all policies assigned to them.
b. Applies filters on their assigned policies (category, due date, status).
c. Accepts policies after passing the quiz (handled in UI).
d. Views compliance records of employees under their supervision with filtering options.

3. Employee (Self Compliance)

a. Views all policies assigned to them.
b. Filters assigned policies by category, due date, or status.
c. Attempts the quiz for each policy (evaluated in UI).
d. Accepts the policy when quiz score ≥ 75%.
e. System updates the compliance status and confirmation date in the database.

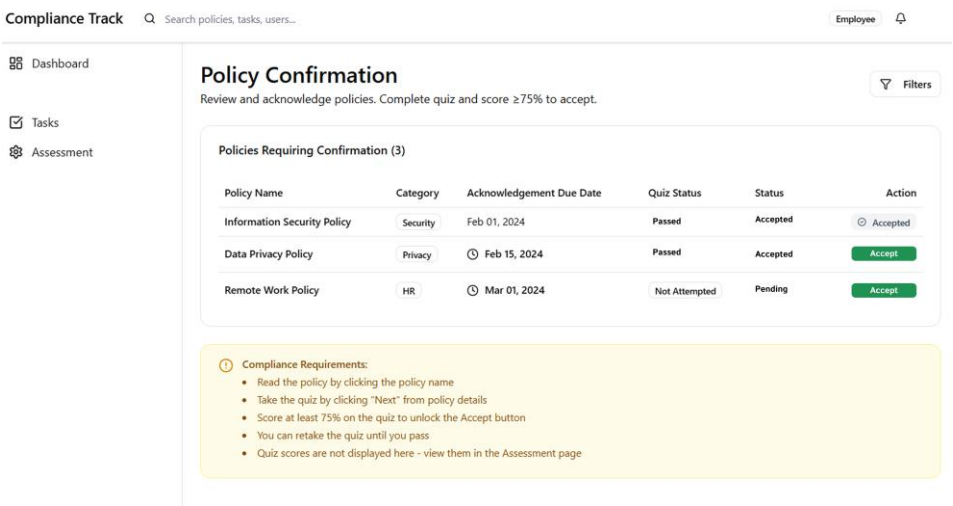
7) ENDPOINTS

Sl.No	METHOD	URL	DESCRIPTION
1	POST	/policyCompliance/assignPoliciesToUser/{policyId}	Assigns the specified policy to all eligible users.
2	DELETE	/policyCompliance/deleteAssignedPolicyToUser/{policyId}/{userId}	Deletes a specific compliance record (policy assignment) for a user.
3	PATCH	/policyCompliance/acceptPolicy	Updates policy acknowledgment status and the confirmation date when employee accepts a policy.
4	GET	/policyCompliance/{userId}/viewAllPolicies	Retrieves the logged-in user's (specified userId) policies with optional filters.
5	GET	/policyCompliance/{managerId}/viewComplianceRecord	Retrieves compliance records of

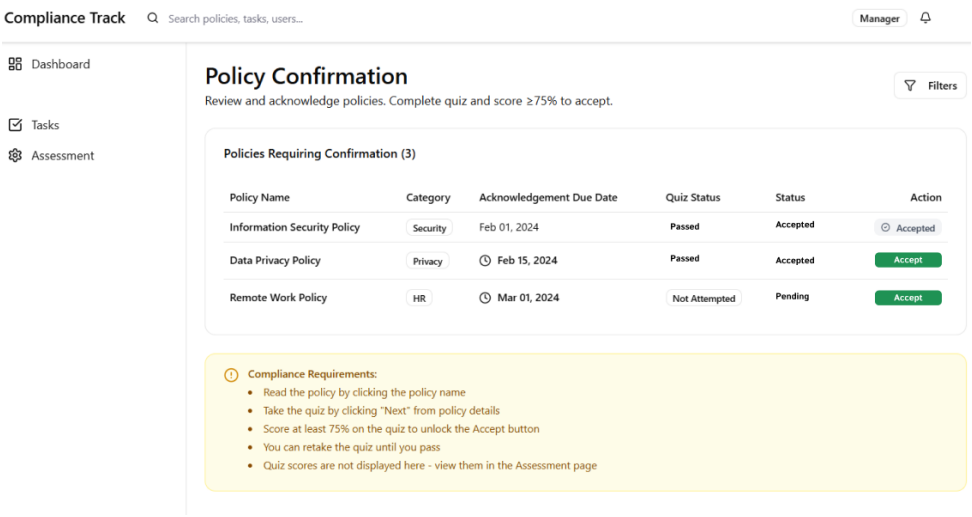
			employees under a manager, with optional filters.
6	GET	/policyCompliance/viewAllComplianceRecords	Retrieves all compliance records for admin dashboard with optional filters.
7	GET	/policyCompliance/searchUsers	Searches compliance records by user name (admin dashboard).

8) SCREENS

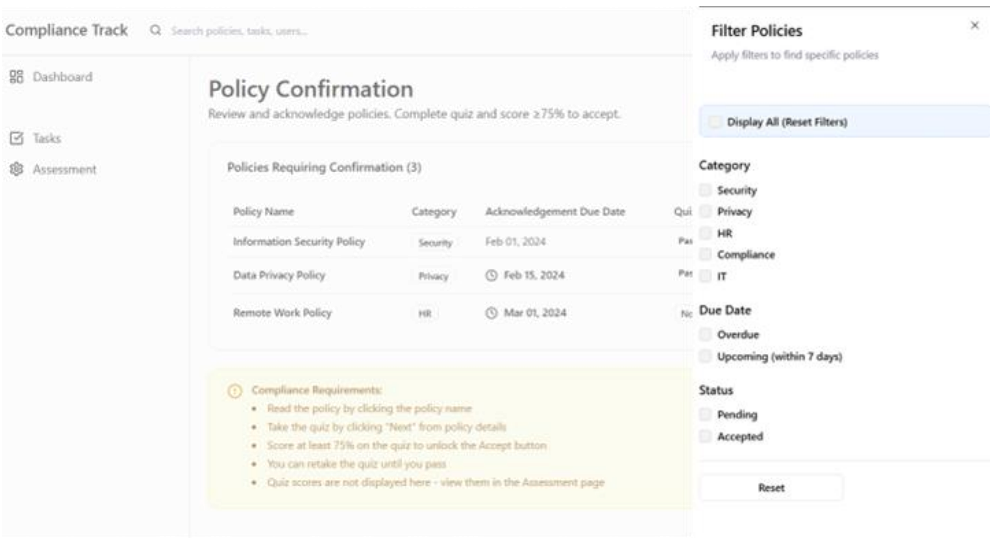
8.1 Employee Policy Confirmation View



8.2 Manager Policy Confirmation View



8.3 Manager Policy Confirmation View



POLICY CATEGORY MANAGEMENT (CRUD)

1) MODULE DESCRIPTION

The Policy Category Management module allows administrators to create, view, update, and delete policy categories that organize corporate policies into a structured classification system. By maintaining clear and consistent categories, the module helps streamline policy grouping, improve searchability, and support effective compliance tracking across the organization.

2) KEY OBJECTIVES

- To enable administrators to create and maintain standardized policy categories.
- To provide a structured way to group multiple policies under relevant categories.
- To ensure consistent and uniform classification of policies across the organization.

3) POLICY CATEGORIES:

The system supports the following predefined corporate policy categories:

- HR and Workplace Behavior Policies
- Ethics and Compliance Policies
- Information Security Policies
- IT Usage and Acceptable Use Policies
- Workplace Health Safety & Environment
- Legal and Corporate Compliance Policies
- Remote Work Attendance and Work Hours Policies

4)DATA MODEL

POLICY_CATEGORY

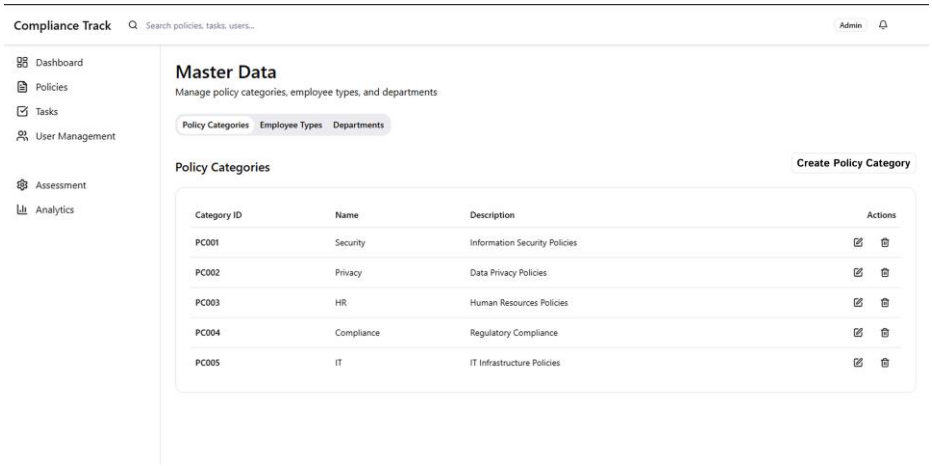
- categoryId
- categoryName|
- categoryDescription

5) ENDPOINTS

Sl.No	METHOD	URL	DESCRIPTION
1	POST	/categories/createPolicyCategory	Creates a new policy category
2	GET	/categories/viewAllPolicyCategory	Retrieves all policy categories
3	GET	/categories/viewPolicyCategory/{categoryId}	Retrieves the description of a

			specific policy category
4	PUT	/categories/{categoryId}/editPolicyCategory	Updates an existing policy category
5	DELETE	/categories/{categoryId}/deletePolicyCategory	Deletes a policy category

6) SCREENS



EMPLOYEE TYPE MANAGEMENT(CRUD)

1)MODULE DESCRIPTION

This module is responsible for performing CRUD operations on employee type . Admin is responsible for performing the crud operations on the employee type.

2)KEY OBJECTIVES

The key objectives of this module are:

- Create new employee types for classification purposes
- Update existing employee type records
- Delete outdated or unused employee types
- Retrieve details of a specific employee type

- Retrieve a paginated and searchable list of all employee types

3)DATA MODEL

EMPLOYEE_TYPE

- empTypeId
- empTypeName

4)ENDPOINTS

Sl. No.	Method	URL	Description
1	POST	/employeeType/create	Creates a new employee type
2	GET	/employeeType/list	Returns paginated & searchable list of employee types.
3	PUT	/employeeType/update	Updates an existing employee type
4	DELETE	/employeeType/delete/{id}	Deletes an employee type by ID.

5)SCREENS

Compliance Track

Search policies, tasks, users...

Admin

- Dashboard
- Policies
- Tasks
- User Management
- Assessment
- Analytics

Master Data

Manage policy categories, employee types, and departments

Policy Categories | Employee Types | Departments

Employee Types

Create

Employee Type ID	Name	Actions
ET001	FTE	
ET002	PTE	
ET003	Intern	
ET004	Temporary	
ET005	Consultant	
ET006	Contractor	

DEPARTMENT MANAGEMENT(CRUD)

1)MODULE DESCRIPTION

This module is responsible for performing CRUD operations on department type . Admin is responsible for performing the crud operations on the department type.

2)KEY OBJECTIVES

The key objectives of this module are:

- To create , update , delete , and read the department type.

3)DATA MODEL

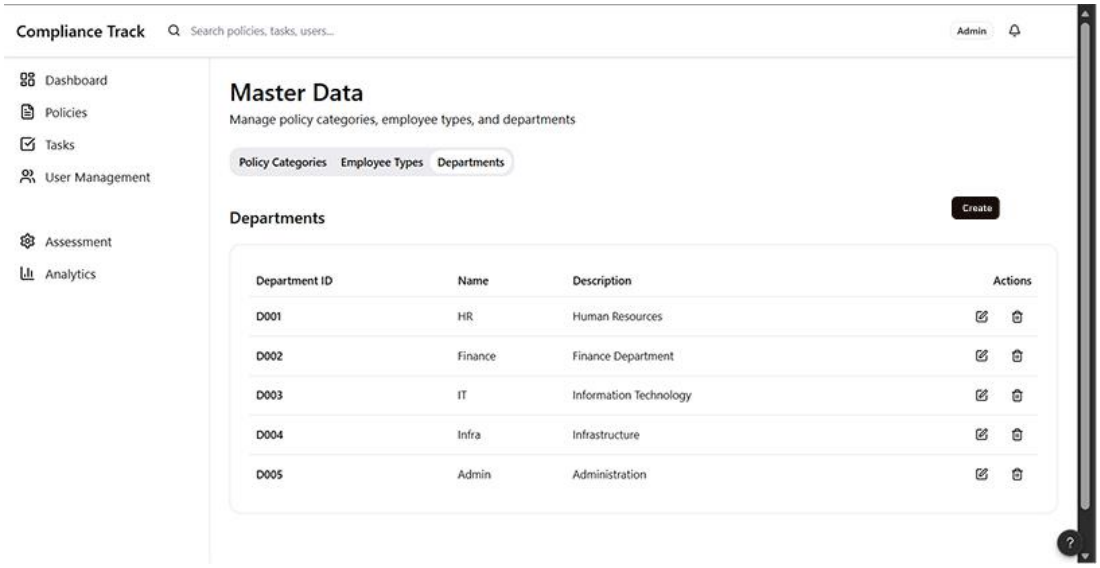
DEPARTMENT

- depId
- depName
- description

4)ENDPOINTS

Sr. No.	Method	URL	Description
1	GET	/ department _type/{id}	Fetches department type of specific id
2	POST	/ department _type/create	Creates new department _type
3	PUT	/department _type/update	Updates the existing department _type
4	DELETE	/department_type/delete/{id}	Deletes the department _type of specific id

5)SCREENS



AUDIT & CHECKLIST MANAGEMENT MODULE

1)MODULE DESCRIPTION

The **Audit & Checklist Management Module** enables consistent, policy-aligned audit execution within the ComplianceTrack system by using a master **AuditChecklist** per policy, from which managers create employee-specific **AuditTasks** populated with actionable **ChecklistItems**. Administrators maintain the reusable checklist templates for each policy, while managers assign tasks, select or copy relevant checklist items, and monitor overall progress. Employees execute their assigned checklist items and mark them as completed, allowing the system to automatically calculate task status—Pending, InProgress, or Completed—based on the proportion of completed items.

2) KEY OBJECTIVES

1. **Model audits consistently**: Use **AuditChecklist** as the *master* library per policy; **AuditTask** as the *employee-specific audit instance*; **ChecklistItem** as the *assigned items under that task*.
2. **Assign & execute**: Admin creates AuditCheckList and Managers create/choose an AuditTask for an employee and populate it with items (copied/selected from AuditChecklist). Employees complete items.
3. **Auto compute progress**: AuditTask progress derives from its ChecklistItems (per employee):
   a. **Pending**: none completed
   b. **InProgress**: some completed
   c. **Completed**: all completed

3) IN SCOPE

- **AuditChecklist** management (Admin): create/edit master checklist **per policy**
- **AuditTask** management (Manager): create/edit per-employee audit instances
- **ChecklistItem** management (Manager creates for employee under AuditTask;.
- Employee updates status.

4) DATA MODEL

A) AuditChecklist

- ItemID
- PolicyID
- Description
- CreatedDate

B) AuditTask

- TaskID
- PolicyID
- AssignedToUserID
- DueDate
- Status (Pending / InProgress / Completed)

C) ChecklistItem

- ItemID
- TaskID
- Description
- Status (Pending / Completed)

5) ACTORS AND USE CASES

5.1 Admin

- Create new **AuditChecklist** items for a specific policy.
- Edit existing **AuditChecklist** items to update descriptions or correct details.
- Delete AuditChecklist items that are outdated or no longer required.
- View all **AuditTasks** created under each policy for monitoring purposes.
- Filter out the policies based in Employee Type and Department type

5.2 Manager

- Create new **AuditTasks** for employees under a specific policy.
- Update existing **AuditTasks**, including due dates or assigned employees.

- Delete AuditTasks that are no longer required or created in error.
- Populate **ChecklistItems** by selecting items from the master **AuditChecklist**.
- Reassign or remove ChecklistItems from an existing AuditTask.
- Track the task's real-time status (Pending, InProgress, Completed) based on employee progress.

5.3 Employee

- View all assigned **AuditTasks** along with due dates and current status.
- Open each task to see the list of **ChecklistItems** assigned by the manager.
- Mark individual ChecklistItems as **Completed** after finishing the required work.
- Track overall task progress (Pending, InProgress, Completed) as items are completed.

6)WORKFLOW:

1. **Admin (Master Setup)**
   a. For a PolicyID, Admin creates AuditChecklist items (policy's reusable checklist library).
2. **Manager (Create Assignment Instance)**
   a. Creates an AuditTask for a specific AssignedToUserID (employee) under a PolicyID; set DueDate; Task Status starts as Pending.
   b. Selects one/all AuditChecklist items for that policy and copies them into ChecklistItem under the new TaskID (Status starts Pending).
   c. System notifies Employee of new assignments.
3. **Employee (Execute)**
   a. Views ChecklistItem list under the TaskID; marks each item Completed when done.
   b. System recalculates AuditTask.Status:
      i. 0 items completed → Pending
      ii. some, not all completed → InProgress
      iii. all completed → Completed

7) ENDPOINTS (BY ROLE)

A) Admin Dashboard

HTTP	Endpoint	Purpose
GET	/admin/policies/{policyId}/audit-checklist	List **AuditChecklist** items for a policy
GET	/admin/policies?employeeType=&departmentType	Filters out the policies based on Employee Type and Department type

POST	/admin/policies/{policyId}/audit-checklist	Create a new **AuditChecklist** item
PUT	/admin/audit-checklist/{itemId}	Update an **AuditChecklist** item
DELETE	/admin/audit-checklist/{itemId}	Delete an **AuditChecklist** item
GET	/admin/audit-tasks?policyId={pid}	View all **AuditTask** under a policy (monitoring)
GET	/admin/notifications	View notifications (e.g., task created)

B) Manager Dashboard

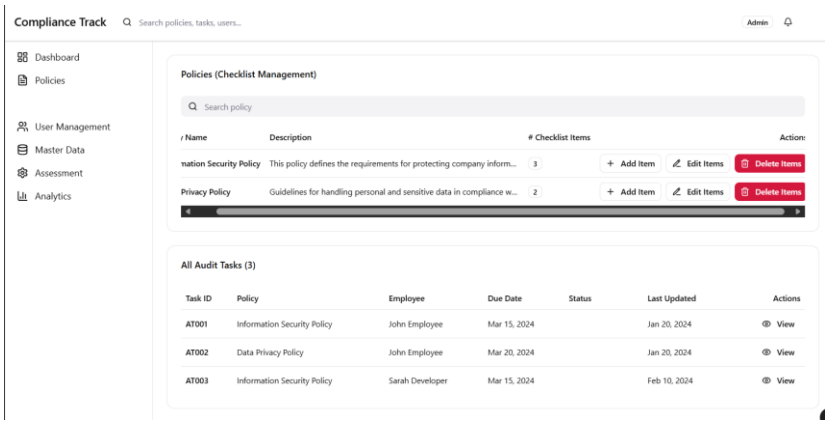
HTTP	**Endpoint**	**Purpose / When it's called**
GET	/manager/employees?employeeType=&departmentType	Populate the **Employee** dropdown with only the employees from particular type and from a particular department.
GET	/manager/policies	Populate the **Policy** dropdown with audit-enabled policies visible to the manager.
GET	/manager/policies/{policyId}/audit-checklist	After the manager picks a policy, load the **master checklist** items for that policy for selection.
POST	/manager/audit-tasks_create-with-items	On **Assign** button click: **create an AuditTask** for the selected **employee + policy + dueDate** and **create ChecklistItem rows** from the selected master checklist items **in one transaction**. Returns Task summary + counts.

C) Employee Dashboard

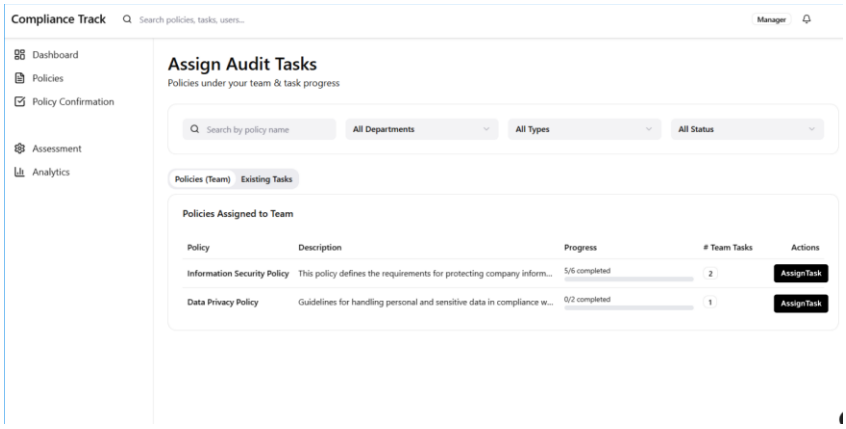
HTTP	Endpoint	Purpose
GET	/employee/audit-tasks	List my **AuditTask** cards with status & due dates
GET	/employee/audit-tasks/{taskId}/checklist-items	View my assigned **ChecklistItem** under the task
PUT	/employee/checklist-items/{itemId}/done	Mark item **Completed**
GET	/employee/audit-tasks/{taskId}/progress	See my task progress (Pending/InProgress/Completed)
GET	/employee/notifications	New assignments, updates

8) SCREENS

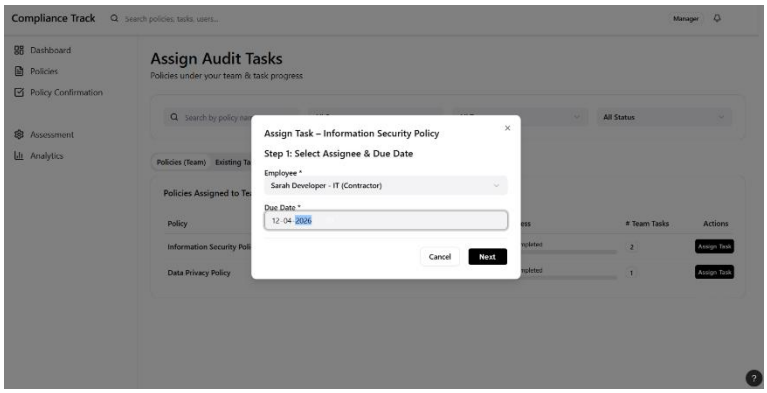
8.1 Admin Audit Task Checklist CRUD Operation



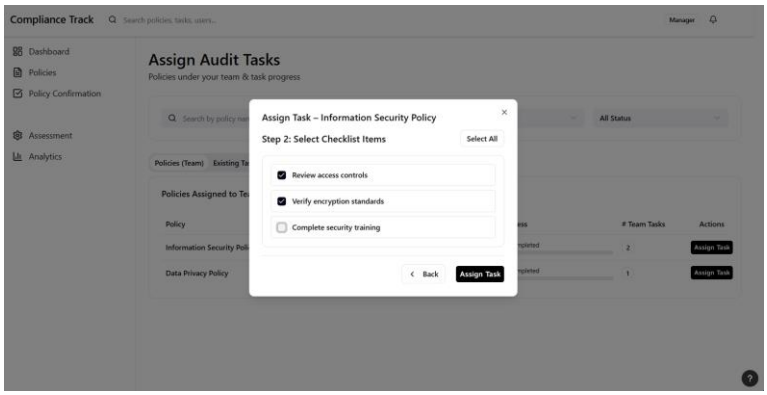
8.2 Manager Audit Task dashboard



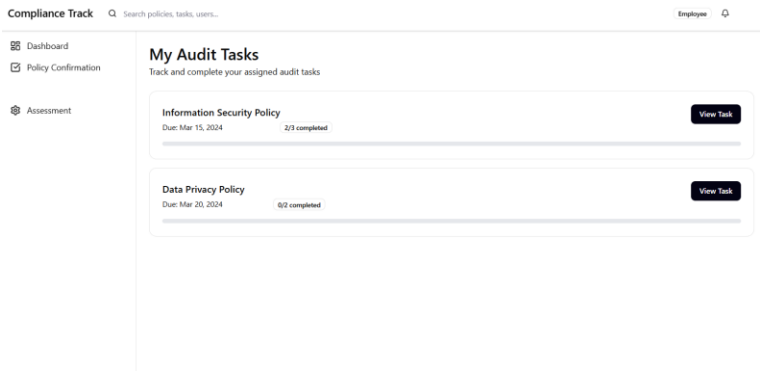
8.3 Manager Assigning Audit Task



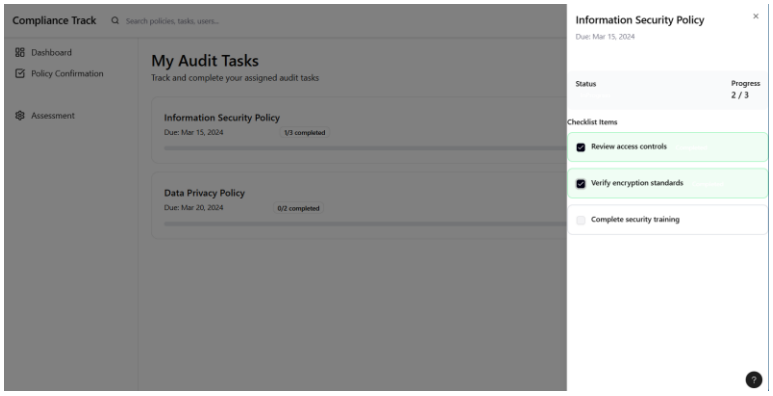
8.4 Checking the tasks to be assigned



8.5 Employee viewing different tasks assigned to him



8.6 Employee marking the tasks he has completed



ANALYTICS & REPORTING MODULE

1) MODULE DESCRIPTION

The Analytics & Reporting Module provides visual insights and summarized compliance performance metrics for administrators and managers.
Admins access organization-wide analytics, while managers receive team-scoped insights.

The module also includes a reporting system that allows users to save analytical snapshots for future viewing.

2)KEY OBJECTIVES

- Provide dashboard KPIs for compliance and audit health
- Support multiple visual charts (bar, donut, pie, histogram, area, bubble)
- Allow managers to view team-only insights
- Enable storage and retrieval of analytical report snapshots
- Ensure strict RBAC-controlled data visibility

3)IN SCOPE

- Dashboard summary metrics
- Policy analytics
- Quiz analytics
- Audit analytics
- Department-wise compliance analytics
- Monthly policy rollout metrics
- Checklist item distribution
- Team pending policies and team performance rankings
- Reports CRUD

4)DATA MODEL

REPORT

- reportId
- title
- metrics
- createdByUserId
- createdAt

5)ACTORS AND USECASES

5.1 Admin

- View organization-wide summary metrics
- View all policy-related analytics
- View audit and compliance analytics
- View department-level compliance
- Save, list, view, rename, delete reports

5.2 Manager

- View team-level summary metrics
- View team policy distribution
- View team audit task status
- View team quiz histogram
- View team pending policies
- View team top performers
- Save, view, rename, delete their own reports

6) WORKFLOW

6.1 Analytics Workflow

1. User opens dashboard
2. System identifies role (Admin/Manager)
3. Displays role-based charts.

Charts provided:

- Most assigned policies (bar)
- Policies by category (horizontal bar)
- Policies with vs without quiz (pie)
- Average quiz score by policy (column)
- Audit task status (donut)

- Compliance by department (stacked bar)
- Monthly policy rollout (area)
- Checklist items per policy (bubble)
- Team quiz score distribution (histogram)
- Team pending policies (horizontal bar)
- Team top performers (ranked bar)

6.3 Reports Workflow

Create Report

- User saves any analytics snapshot by assigning or auto-generating a title.
- System stores the metrics and user info.

View Report

- User opens a saved report to see the snapshot as it was at creation time.

List Reports

- Admin sees all reports.
- Manager sees only their own.

Rename Report

- User updates only the title.

Delete Report

- User deletes a report they created.
- Admins may delete any report.

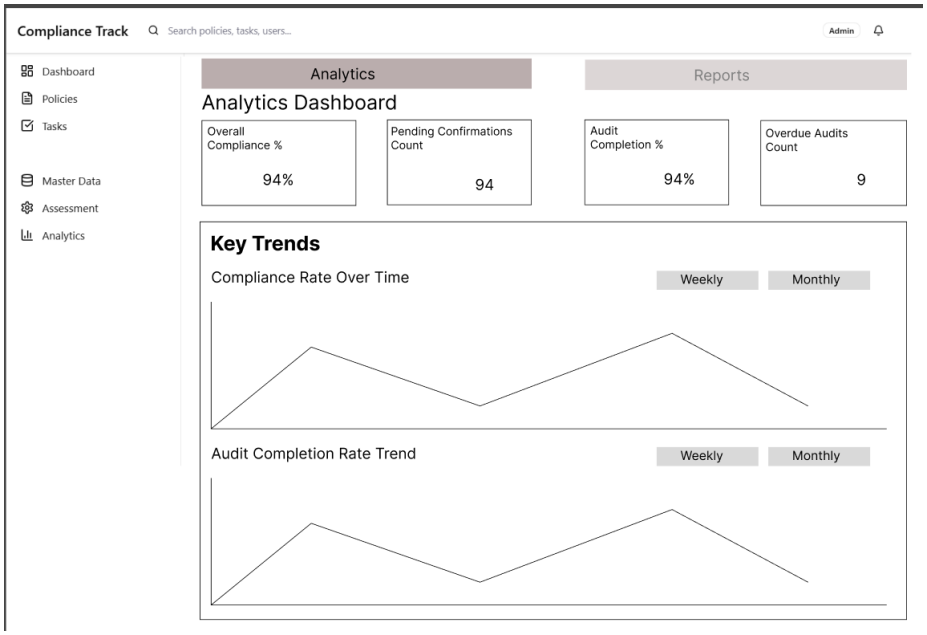
7) ENDPOINTS

Sl. No.	Method	URL	Description
1	GET	/analytics/summary	Returns overall compliance health (overall compliance %, pending confirmations, audit completion %, overdue audits).
2	GET	/analytics/policies/most-assigned	Most assigned policies.

3	GET	/analytics/policies/by-category	Policies grouped by category
4	GET	/analytics/quiz/avg-score-by-policy	Average quiz score per policy
5	GET	/analytics/audit/status	Audit task status distribution (Pending/InProgress/Completed).
6	GET	/analytics/policies/with-vs-without-quiz	Policies with/without quiz.
7	GET	/analytics/compliance/department	Department-wise compliance.
8	GET	/analytics/policies/rollout/monthly	Monthly policy rollout count.
9	GET	/analytics/policies/checklists/bubble	Checklist item count per policy (bubble chart)
10	GET	/analytics/team/quiz-scores/histogram	Team quiz score distribution (histogram)
11	GET	/analytics/team/pending-policies	Team pending policies.
12	GET	/analytics/team/top-performers	Team top performers based on quiz scores.
13	POST	/reports/create	Creates a new saved report.
14	GET	/reports/{id}	Fetch specific saved report
15	GET	/reports	List saved reports (pagination + search)
16	PATCH	/reports/{id}	Rename report
17	DELETE	/reports/{id}	Delete saved report.

8) SCREENS

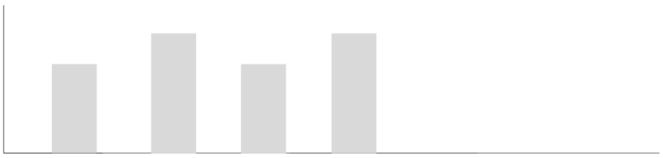
8.1 Analytics



Dashboard
Policies
Tasks

Master Data
Assessment
Analytics

Average Quiz Score by Policy



Policies with Low Compliance

Category	Policy Name	Compliance %
Category	Policy Name	Compliance %
Category	Policy Name	Compliance %
Category	Policy Name	Compliance %

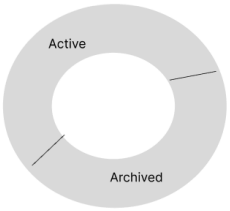
Overdue Audit Tasks

Task Id	Policy Name	Assigned To	Due Date	Status
Task Id	Policy Name	Assigned To	Due Date	Status
Task Id	Policy Name	Assigned To	Due Date	Status
Task Id	Policy Name	Assigned To	Due Date	Status

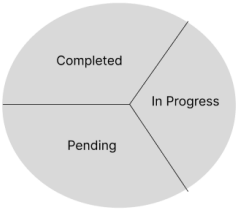
Dashboard
Policies
Tasks

Master Data
Assessment
Analytics

Policy Status Distribution



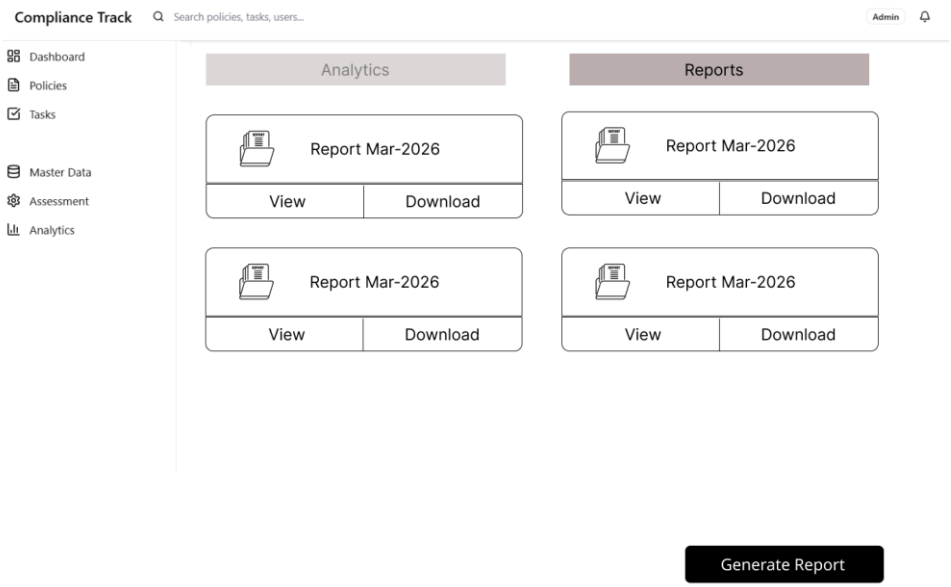
Audit Status Distribution

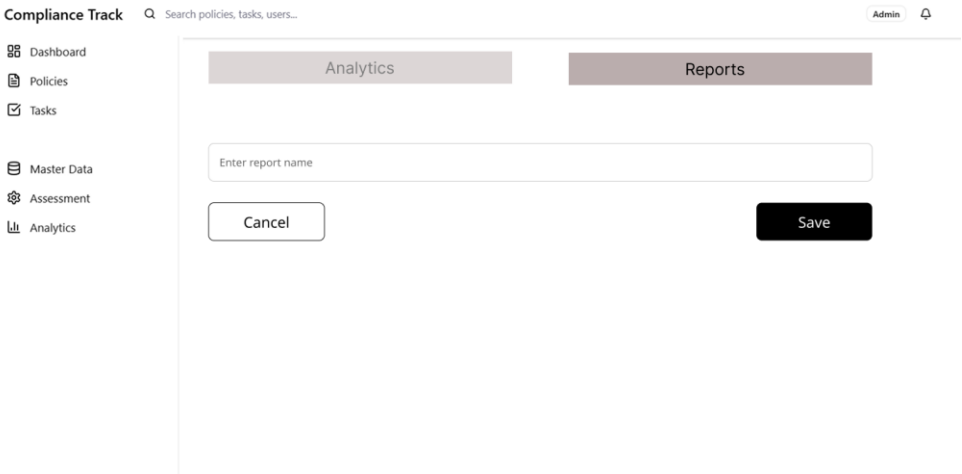


Team Compliance % grouped by Manager

Scroll to Top

8.2 Reports


Compliance Track
Search policies, tasks, users...
Admin
Dashboard
Policies
Tasks
Master Data
Assessment
Analytics
Analytics
Reports
Report Mar-2026
View
Download
Report Mar-2026
View
Download
Report Mar-2026
View
Download
Report Mar-2026
View
Download
Generate Report


Compliance Track
Search policies, tasks, users...
Admin
Dashboard
Policies
Tasks
Master Data
Assessment
Analytics
Analytics
Reports
Enter report name
Cancel
Save

NOTIFICATIONS & IN-APP ALERTS MODULE

1)MODULE DESCRIPTION

The Notifications & In-App Alerts Module is responsible for informing users about important system events related to compliance confirmations, audit tasks, and deadlines.

2)KEY OBJECTIVES

The key objectives of this module are:

- To alert employees about **pending compliance confirmations**
- To notify users about **assigned audit tasks and approaching deadlines**
- To provide **realtime visibility** of unread alerts inside the application
- The system supports the following notification types:

Notification Type	**Description**
ComplianceReminder	Triggered when an employee has not confirmed a policy
AuditAssignment	Triggered when an audit task is assigned
AuditDeadline	Triggered as audit due date approaches
AuditOverdue	Triggered when an audit task is overdue
PolicyUpdate	Triggered when a new or updated policy becomes active

3)IN SCOPE

- Inapp notification delivery and UI
- Targeted notifications (userspecific) and admin **broadcasts**
- Filtering (type, status, date range, search), pagination, sorting
- Read/unread state, unread count
- Admin Broadcast Center (create/schedule/send) // optional

4)DATA MODEL

NOTIFICATION

- notificationId
- userId
- notificationType

- title
- ntificationSeverity
- message
- status
- createdAt

5)ACTORS AND USE CASES:

5.1 **Employee**

- See notifications on login for **pending confirmations**, **audit deadlines/overdues**
- Filter notifications and act from deep links
- Mark notifications as read

5.2 Manager

- Receive **PolicyUpdate** alerts on new/updated policies
- Optionally see **team audit deadlines** (future enhancement)
- Filter by type and status

5.3 Admin

- **Broadcast** orgwide or scoped messages (roles/departments)

6)WORKFLOW

- Policy published/updated → **PolicyUpdate** (Managers; employees if requires confirmation)
- Compliance pending → **ComplianceReminder**
- Audit task assigned → **AuditAssignment**
- Due soon → **AuditDeadline**
- Overdue → **AuditOverdue**
- Admin announcements → **Broadcast**

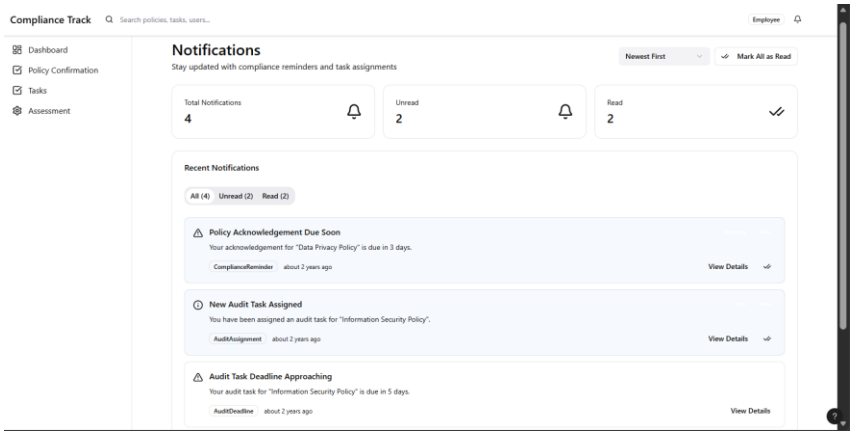
7)ENDPOINTS

Sl. No.	Method	URL	Description
1	GET	/notifications?status=&type=&search=&sort=&page=&size=	Fetches a paginated list of notifications based on optional filters such as status,

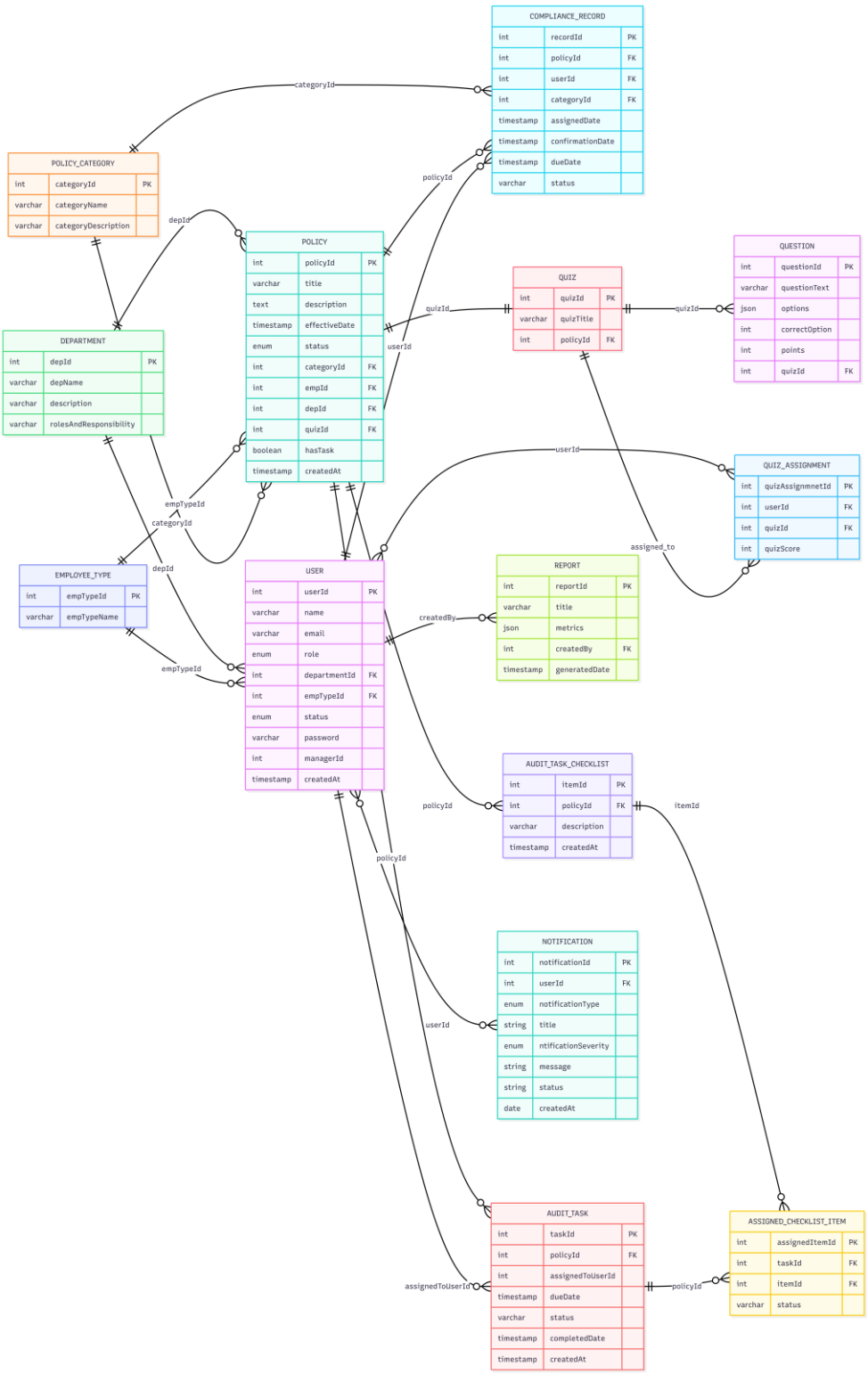
			type, search query, sorting criteria, page number, and page size.
2	GET	notifications/unread/count	Retrieves the count of unread notifications for the user.
3	PATCH	notifications/{id}/read	Marks a specific notification (identified by its ID) as read.
4	PATCH	notifications/read-all	Marks all notifications for the user as read in a single operation.

8)SCREENS

8.1 Notification screen



ER DIAGRAM


COMPLIANCE_RECORD
int recordId PK
int policyId FK
int userId FK
int categoryId FK
timestamp assignedDate
timestamp confirmationDate
timestamp dueDate
varchar status
POLICY_CATEGORY
int categoryId PK
varchar categoryName
varchar categoryDescription
POLICY
int policyId PK
varchar title
text description
timestamp effectiveDate
enum status
int categoryId FK
int empId FK
int depId FK
int quizId FK
boolean hasTask
timestamp createdAt
QUIZ
int quizId PK
varchar quizTitle
int policyId FK
QUESTION
int questionId PK
varchar questionText
json options
int correctOption
int points
int quizId FK
DEPARTMENT
int depId PK
varchar depName
varchar description
varchar rolesAndResponsibility
QUIZ_ASSIGNMENT
int quizAssignmnetId PK
int userId FK
int quizId FK
int quizScore
EMPLOYEE_TYPE
int empTypeId PK
varchar empTypeName
USER
int userId PK
varchar name
varchar email
enum role
int departmentId FK
int empTypeId FK
enum status
varchar password
int managerId
timestamp createdAt
REPORT
int reportId PK
varchar title
json metrics
int createdBy FK
timestamp generatedDate
AUDIT_TASK_CHECKLIST
int itemId PK
int policyId FK
varchar description
timestamp createdAt
NOTIFICATION
int notificationId PK
int userId FK
enum notificationType
string title
enum ntificationSeverity
string message
string status
date createdAt
AUDIT_TASK
int taskId PK
int policyId FK
int assignedToUserId
timestamp dueDate
varchar status
timestamp completedDate
timestamp createdAt
ASSIGNED_CHECKLIST_ITEM
int assignedItemId PK
int taskId FK
int itemId FK
varchar status
categoryId
policyId
depId
quizId
quizId
userId
empTypeId
categoryId
depId
empTypeId
userId
assigned_to
createdBy
policyId
itemId
policyId
userId
assignedToUserId
policyId